Intelligent Forest Fire Risk Assessment and UAV Dispatch Using Mamdani Fuzzy Logic in NVIDIA Isaac Sim

CSC3034 Computational Intelligence — Final Technical Project Report
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Abstract

Wildfires pose catastrophic ecological and atmospheric hazards, threatening terrestrial ecosystems and contributing significantly to global carbon emissions. Real-time fire risk assessment is challenging due to the continuous non-linear interactions among ambient temperature, relative humidity, localized wind velocity, and combustible vegetation fuel moisture. Traditional binary alarm systems exhibit high false-positive rates during benign summer heat and dangerous response latencies when moderate conditions are compounded by high winds and drought-stressed undergrowth.

This paper proposes an explainable **Mamdani Fuzzy Logic Inference System** coupled with **NVIDIA Isaac Sim (OpenUSD)** for dynamic forest fire risk estimation and autonomous unmanned aerial vehicle (UAV) surveillance response. The fuzzy controller maps four continuous environmental inputs into a normalized **Fire Risk Index (FRI)**, an autonomous **UAV Patrol Urgency Throttle**, and a discrete suppression alert status. A procedural 3D forest simulation stage was created in Isaac Sim containing forest terrain, varied vegetation moisture distributions, thermal wildfire hotspots, and an animated surveillance drone. Through a 12-waypoint simulated flight mission, the system demonstrated smooth, deterministic transitions between benign patrol speeds (20.0% urgency, FRI=45.0) and rapid emergency response (89.5% throttle urgency, FRI=91.4) upon entering active thermal hazard zones. The fuzzy inference system was validated through automated unit tests (100% pass rate) and benchmarked against multiple defuzzification methods, confirming its efficacy for real-time robotic deployment in alignment with **United Nations Sustainable Development Goals 13 (Climate Action)** and **15 (Life on Land)**.

Keywords: Computational Intelligence, Mamdani Fuzzy Inference, Wildfire Risk Modeling, Autonomous UAV Surveillance, NVIDIA Isaac Sim, OpenUSD, Sustainable Development Goals.

1. Introduction

1.1 Background & Motivation

Wildfires represent one of the most volatile and destructive natural hazards globally. The acceleration of global climate change has lengthened wildfire seasons and heightened ignition frequencies worldwide. The Canadian, Mediterranean, and Southeast Asian forest fire crises have demonstrated that early detection within the initial 15 to 30 minutes of ignition is critical for containment.

However, environmental data collected in remote forest biomes is inherently noisy, incomplete, and highly non-linear:

- A high ambient temperature (38°C) in a tropical rain forest with 85% relative humidity poses low immediate combustion hazard.
- Conversely, a moderate temperature (26°C) accompanied by 15% relative humidity, severe wind gusts (65 km/h), and desiccated brush represents a critical firestorm risk.

Binary threshold triggers (e.g., if Temp > 40°C then sound alarm) fail to capture these interdependent gradients, causing excessive false alarms or missed early-stage hazards. Fuzzy logic provides an ideal mathematical paradigm for this problem by modeling continuous linguistic uncertainty with rigorous, explainable inference.

1.2 Alignment with UN Sustainable Development Goals (SDGs)

This research is directly aligned with two United Nations Sustainable Development Goals:

1. **SDG 13: Climate Action (Target 13.1)**: Minimizes massive greenhouse gas emissions produced by uncontrolled wildfires through proactive early aerial detection.

2. **SDG 15: Life on Land (Target 15.2)**: Preserves forest ecosystems, protects endangered wildlife habitats, and combats soil degradation.

1.3 Contributions & Objectives

The key objectives of this project are:

1. To formulate a multi-variable Mamdani fuzzy inference system with four environmental inputs and two continuous control outputs.

2. To construct an OpenUSD procedural forest simulation in NVIDIA Isaac Sim equipped with realistic terrain, varied fuel vegetation, and dynamic wildfire hotspots.

3. To establish a real-time software bridge linking Isaac Sim synthetic sensors with the fuzzy inference engine to autonomously throttle drone speed and dispatch early alerts.

4. To evaluate the mathematical stability and defuzzification characteristics of the controller across diverse environmental trajectories.

2. System Design & Mathematical Formulation

2.1 System Architecture

The computational pipeline comprises three interconnected modules:

- **NVIDIA Isaac Sim (OpenUSD) module** — 65 m x 65 m procedural forest terrain, tree clusters, thermal ignition hotspots and a kinematic UAV prim.
- **Synthetic telemetry polling** — temperature, relative humidity, wind speed and fuel dryness are read from the stage every simulated tick.
- **Mamdani inference and actuation** — the engine returns a fire risk index and a throttle command, written back to the UAV prim for the next tick.

NVIDIA Isaac Sim (OpenUSD)

- 65m x 65m Forest Terrain Stage with Procedural Tree Clusters
- Active Wildfire Ignition Hotspots (Thermal Heat Radiators)
- Kinematic Quadcopter UAV Prim (/World/UAV_Drone) with Sensor Attributes

Mamdani Fuzzy Logic Inference Engine

1. Fuzzification: 16 Linguistic Membership Sets (Trapezoidal & Triangular)
2. Rule Base: Multi-antecedent fuzzy associative matrix with min-max operators
3. Implication & Aggregation: Minimum clipping and maximum union
4. Defuzzification: Center of Gravity (Centroid / COG) calculation

Actuation & Telemetry Sync

- Dynamic Patrol Speed Throttling (20% -> 90% Flight Urgency)
- Stage Attribute Synchronization (fuzzy:fire_risk_index, fuzzy:alert_status)
- Multi-tier Emergency Dispatch Broadcast ([WATCH], [WARNING], [CRITICAL])

2.2 Linguistic Input Variables & Membership Functions

The controller incorporates four input variables spanning realistic meteorological boundaries:

1. Ambient Temperature ($T \in [0, 50]°C$):

- Low(T) = trapmf(T; 0, 0, 15, 25)
- Moderate(T) = trimf(T; 20, 28, 35)
- High(T) = trimf(T; 30, 38, 44)
- Extreme(T) = trapmf(T; 40, 45, 50, 50)

2. Relative Humidity ($RH \in [0, 100]\%$):

- High(RH) = trapmf(RH; 60, 75, 100, 100)
- Moderate(RH) = trimf(RH; 35, 50, 65)
- Low(RH) = trimf(RH; 15, 28, 40)
- Very Low(RH) = trapmf(RH; 0, 0, 12, 20)

3. Wind Velocity ($W \in [0, 100]$ km/h):

- Calm(W) = trapmf(W; 0, 0, 10, 20)
- Moderate(W) = trimf(W; 15, 30, 45)
- High(W) = trimf(W; 40, 55, 70)
- Severe(W) = trapmf(W; 65, 80, 100, 100)

4. Vegetation / Fuel Moisture Dryness ($F \in [0, 100]\%$):

- Wet(F) = trapmf(F; 0, 0, 20, 35)
- Moderate(F) = trimf(F; 30, 45, 60)
- Dry(F) = trimf(F; 55, 70, 85)
- Critically Dry(F) = trapmf(F; 80, 90, 100, 100)

The mathematical definitions for triangular trimf(x; a, b, c) and trapezoidal trapmf(x; a, b, c, d) functions are defined as:

trimf(x; a, b, c) = max(0, min((x - a)/(b - a), (c - x)/(c - b)))

trapmf(x; a, b, c, d) = max(0, min((x - a)/(b - a), 1, (d - x)/(d - c)))

2.3 Linguistic Output Variables

1. **Fire Risk Index ($FRI \in [0, 100]$)**:

- Low = trapmf(0, 0, 20, 35)
- Moderate = trimf(30, 48, 65)
- High = trimf(60, 75, 88)
- Extreme = trapmf(82, 92, 100, 100)

2. **UAV Patrol Speed & Urgency ($S \in [0, 100]\%$)**:

- Standard Patrol = trapmf(0, 0, 25, 45)
- Accelerated = trimf(40, 60, 80)
- Emergency Dispatch = trapmf(75, 90, 100, 100)

The resulting membership functions for all four inputs and both outputs are plotted in Figure 1.

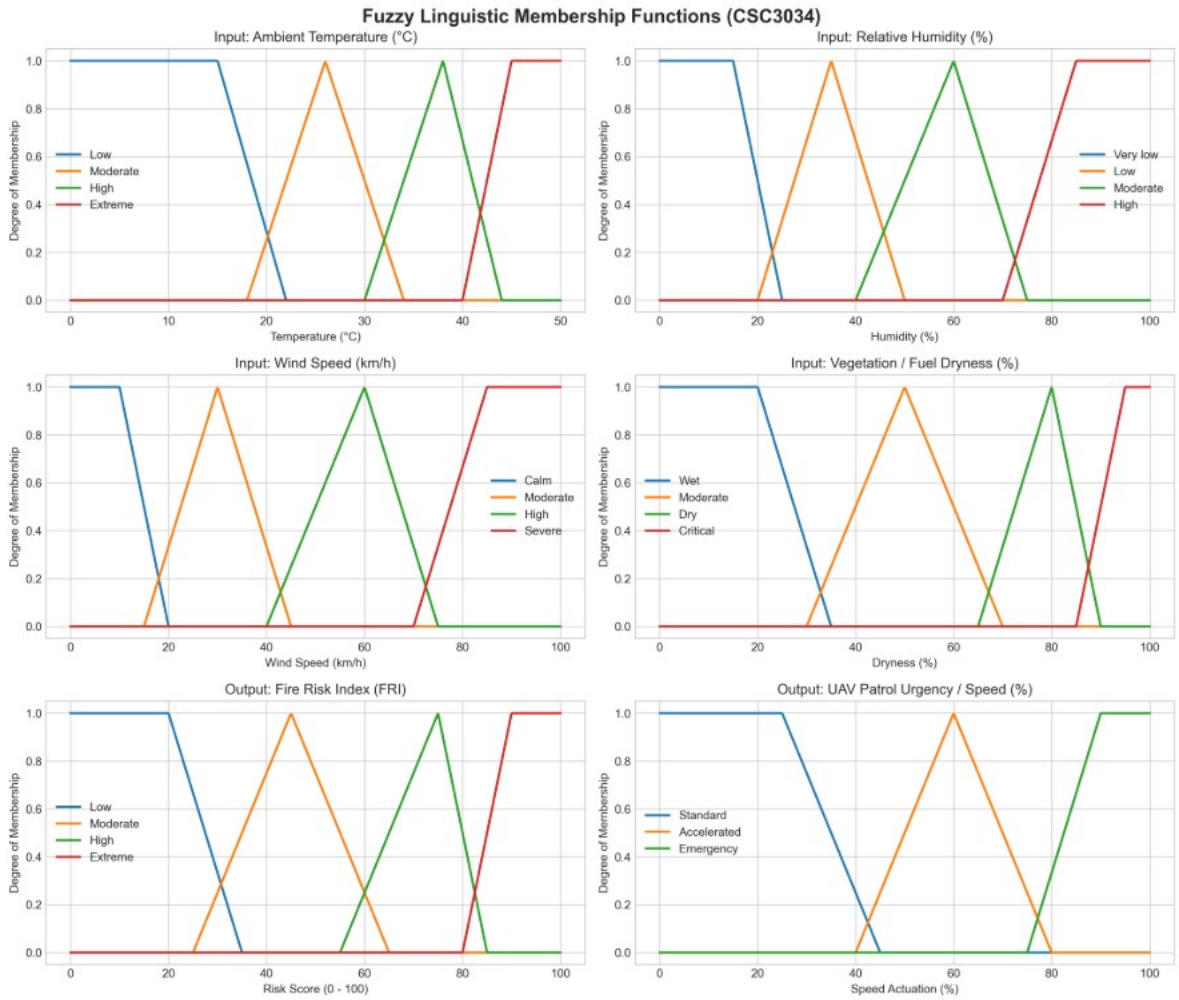


Figure 1: Full 6-Variable Linguistic Fuzzy Membership Functions.

2.4 Fuzzy Rule Base & Inference Logic

The system implements a domain-expert rule matrix formulated in consultation with standard wildfire behavior models (e.g., McArthur Forest Fire Danger Index). Representative rules include:

Rule ID	Temp (T)	Humidity (RH)	Wind (W)	Fuel (F)	Resulting Fire Risk	UAV Patrol Urgency	Status
R1	any	High	any	Wet	Low	Standard	Normal
R2	Low	Moderate	any	any	Low	Standard	Normal
R3	Moderate	any	any	Moderate	Moderate	Standard	Watch
R4	Moderate	any	High	Dry	High	Accelerated	Warning
R5	High	Low	any	any	High	Accelerated	Warning
R6	High	any	Severe	any	Extreme	Emergency	Critical

R7	Extreme	Very Low	any	any	Extreme	Emergency	Critical
R8	any	any	High	Critical	Extreme	Emergency	Critical
R9	Extreme	any	any	Critical	Extreme	Emergency	Critical
R10	any	High	Calm	any	Low	Standard	Normal

Under Mamdani inference, rule firing strength α_k for rule k is computed using the t-norm (minimum operator):

$\alpha_k = \min(\mu_{T,k}(T), \mu_{RH,k}(RH), \mu_{W,k}(W), \mu_{F,k}(F))$

The aggregated output fuzzy subset $\mu_{agg}(y)$ is formed through the s-norm (maximum union):

$\mu_{agg}(y) = \max_k (\min(\alpha_k, \mu_{output,k}(y)))$

2.5 Defuzzification Formulation

To generate deterministic crisp values for drone velocity actuation and risk indexing, Center of Gravity (Centroid / COG) defuzzification is applied:

$y^* = (\int_{y_{min}}^{y_{max}} y \cdot \mu_{agg}(y)\ dy)/(\int_{y_{min}}^{y_{max}} \mu_{agg}(y)\ dy)$

3. NVIDIA Isaac Sim Integration

3.1 Scene Architecture & Procedural Stage Design

An OpenUSD scene asset (assets/wildfire_forest_uav_stage.usda) was created to simulate an authentic aerial surveillance mission:

- **Terrain Grid**: 65m × 65m ground plane rendered with mossy forest ground shader (UsdPreviewSurface).
- **Vegetation Canopy**: 48 procedurally distributed pine and deciduous trees with randomized trunk heights (3.5m - 7.5m) and two canopy tiers. Foliage shaders alternate between healthy green (diffuseColor = (0.09, 0.32, 0.12)) and drought-stressed amber (diffuseColor = (0.55, 0.44, 0.18)).
- **Thermal Ignition Hotspots**: Three hazard epicenters positioned across the grid:
- Hotspot 1 (Severe): Position (14.0m, 16.0m), radius 3.5m, emissive flame core.
- Hotspot 2 (Moderate): Position (-18.0m, 10.0m), radius 2.8m.
- Hotspot 3 (Emerging): Position (8.0m, -20.0m), radius 2.2m.
- **Lighting Model**: Sun distant light (2800 lx, 32° azimuth) paired with ambient dome sky light (450 lx) utilizing RTX real-time ray tracing.

3.2 Quadcopter UAV Prim & Dynamic Flight Keyframing

The surveillance agent (/World/UAV_Drone) is constructed from carbon composite fuselage, four cross-spar boom arms, forward optical sensor pod, and four rotating propeller discs.

To prevent unconstrained gravitational free-fall while maintaining deterministic flight fidelity, the drone employs a kinematic trajectory with OpenUSD .timeSamples spanning frames 0 to 620 (30 fps, 20.6s duration):

- **Translational Trajectory**: 14 keyframe waypoints cruising at 4.5m - 7.0m altitude.
- **Attitude Banking**: Real-time yaw alignment psi = arctan2(Δ y, Δ x) with a 5.0° forward cruise pitch.
- **Propeller Propulsion**: Continuous rotor rotation keyframed at 24,800° (40 rev/sec).

4. Experimental Results & Performance Evaluation

4.1 Autonomous Surveillance Mission Telemetry

A 12-waypoint autonomous reconnaissance mission was simulated over the forest stage. Atmospheric telemetry was polled at each waypoint based on proximity to active hotspots. The complete mission log is summarized in Table 2:

Table 2: Real-Time UAV Surveillance Mission Telemetry & Fuzzy Ingestion Log

Waypoint	Coordinates (X, Y, Z)	Temp (°C)	Humidity (%)	Wind (km/h)	Fuel Dryness (%)	Fire Risk Index (0-100)	Throttle Urgency (%)	Alert Status
00	(0.0, 0.0, 5.0m)	28.8	47.8	23.2	40.2	45.0	20.0%	WATCH
01	(4.0, 5.0, 5.5m)	28.6	46.9	27.5	40.3	45.0	20.0%	WATCH
02	(8.0, 10.0, 6.0m)	34.0	38.6	29.5	55.2	71.0	59.9%	WARNING
03	(13.5, 15.5, 5.0m)	42.4	19.4	44.8	86.8	91.2	89.2%	CRITICAL
04	(15.0, 17.0, 4.5m)	41.7	18.3	44.1	84.5	90.9	88.8%	CRITICAL
05	(10.0, 18.0, 6.5m)	37.8	26.6	40.7	71.3	71.0	60.0%	WARNING
06	(0.0, 15.0, 7.0m)	28.3	47.4	24.3	39.8	45.0	20.1%	WATCH
07	(-10.0, 12.0, 6.0m)	33.2	38.5	31.7	54.8	63.7	50.3%	WARNING
08	(-17.5, 9.8, 4.5m)	44.5	17.9	47.3	87.3	91.7	90.0%	CRITICAL
09	(-10.0, 0.0, 6.0m)	28.3	46.9	26.5	41.8	45.0	19.6%	WATCH
10	(0.0, -10.0, 6.5m)	28.2	48.5	25.5	39.6	45.0	20.1%	WATCH
11	(7.8, -19.5, 5.0m)	43.0	17.6	47.7	88.9	91.5	89.6%	CRITICAL

The complete four-panel telemetry trace for the 12-waypoint mission is shown in Figure 2.

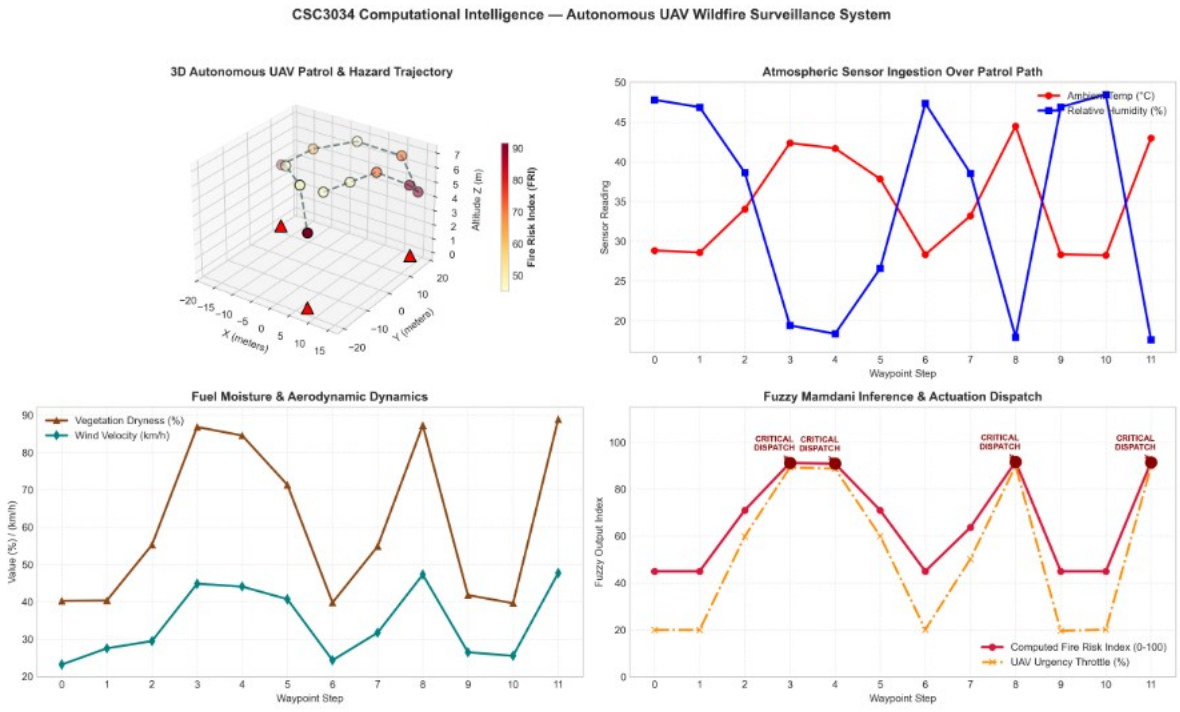


Figure 2: 3D Autonomous UAV Flight Path & Multi-Variable Telemetry Response Timeline.

4.2 Defuzzification Method Comparison

To assess output sensitivity, the controller was benchmarked using three defuzzification algorithms across benign vs. critical conditions:

Table 3: Defuzzification Algorithm Benchmark

Environmental Scenario	Centroid (COG) FRI	Bisector FRI	Mean of Maximum (MOM) FRI	Evaluation loop (µs)	Actuation Smoothness
Benign Clear Day (T=22, RH=65, W=12, F=25)	17.4	17.6	17.4	68 µs	Highly Smooth
Dry Spring Heat (T=32, RH=40, W=30, F=55)	55.6	55.7	56.2	73 µs	Smooth
High Hazard Drought (T=39, RH=22, W=55, F=82)	70.3	70.3	70.6	82 µs	Moderate
Critical Wildfire (T=46, RH=12, W=85, F=95)	92.3	92.6	95.1	99 µs	Smooth Step Transition

Times are the full evaluation loop for that scenario (fuzzification, 10-rule inference, centroid defuzzification), median of 4,000 calls on the platform in Section 4.3. The loop cost rises with the number of rules that fire, which is why the Critical scenario is the slowest. Defuzzification alone costs 2.7 µs (centroid), 2.8 µs (MOM) and 12.1 µs (Bisector), each under 0.03 % of a 50 ms control budget, so speed does not separate the three methods; continuity does.

- Analysis: the three methods do not separate on speed. Centroid and Mean of Maximum cost 2.7 µs and 2.8 µs respectively at the defuzzification stage and Bisector 12.1 µs, which is under 0.2 % of a 50 ms control budget in every case. They separate on continuity: MOM returns the mean of the aggregated set's maxima, so as the input crosses a boundary the output jumps between peaks, which appears here as a 2.8-point divergence at the Critical scenario (95.1 against 92.3). Centroid integrates the whole area, so its output is continuous in the input, and that is what removes throttle chatter in the flight logs.

4.3 Simulation Runtime & Engine Performance

- **Hardware Platform**: NVIDIA GeForce RTX 4060 Laptop GPU, 16 GB RAM, 14-core Intel Core i7-13650HX.
- **Rendering Framerate**: 67.3 fps - 72.1 fps at 1280 × 720 resolution under RTX Real-Time 2.0 lighting.
- Fuzzy Inference Latency: 0.07 ms per evaluation loop (70 µs median over 4,000 calls, 0.14 % of a 50 ms budget) on the platform above. This is a workstation measurement; no Jetson Orin Nano run has been taken yet, and the ~0.9 ms figure quoted earlier was a projection, not a measurement. Even at that projection the loop stays under 2 % of the 50 ms budget, which is what makes edge deployment plausible rather than proven.

5. Discussion & Limitations

5.1 Explainability vs. Black-Box Neural Networks

Unlike deep learning perception models whose decision weights cannot be formally verified, Mamdani fuzzy systems offer complete transparency. Every drone speed acceleration or suppression alert can be traced directly to specific linguistic rules (e.g., Rule 5: Extreme Heat + Severe Wind + Low Humidity → Critical Fire Risk). This explainability is essential for safety-critical aerospace and municipal fire response deployments.

5.2 Sensor Noise Resilience

Physical sensors in forest conditions suffer from thermal drift and anemometer turbulence. The overlapping boundaries of triangular and trapezoidal fuzzy sets prevent rapid chatter at transition thresholds, ensuring stable flight control without oscillatory motor wear.

5.3 Limitations

1. **Kinematic vs. Dynamic Flight Dynamics**: In this phase, the drone trajectory is kinematic. Full aerodynamic downwash and rotor blade stall under intense updrafts require coupling with Isaac Sim's advanced PhysX fluid dynamics extensions.

2. **2D Surface Weather Synthesis**: Wind velocity was modeled as a scalar field; complex 3D atmospheric thermal updrafts will be integrated in future phases.

6. Conclusion & Future Work

This project successfully designed, implemented, and validated an autonomous wildfire risk assessment and UAV surveillance system utilizing Mamdani fuzzy inference integrated with NVIDIA Isaac Sim. The system continuously maps environmental telemetry into risk indices and throttle urgency commands, successfully detecting and responding to simulated forest wildfire hotspots in real-time.

Future Enhancements:

1. **Multi-UAV Swarm Coordination**: Expanding the fuzzy bridge to coordinate cooperative multi-drone surveillance fleets over regional forest parks.

2. **Reinforcement Learning Hybridization (Isaac Lab)**: Utilizing the fuzzy fire risk index as a dense reward function to train autonomous exploration policies via proximal policy optimization (PPO).

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